

SOFTWARE CONSTRUCTION & DEVELOPMENT (SESC-344)

I. Course Details

Credit Hours	3 (2+1)
Pre-requisites	Software Design and Architecture
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	1. Clean Code: A Handbook of Agile Software Craftsmanship, Robert C. Martin, Prentice Hall, 2008. 2. Code Complete 2nd edition: A practical handbook of software construction, published by Microsoft Press, 2004. 3. Object-Oriented Software Construction, by Bertrand Meyer, Second Edition, Published by, Prentice Hall in 1997. 4. Software Engineering by Ian Sommerville, 8th edition, Addison & Wesley. 2006.
Recommended Reference (Books/Websites/Articles)	1. Introduction To Computer Theory Daniel I. A. Cohen, Wiley, 17-Jan-1986. 2. Teach Yourself Uml In 24 Hours, Joseph Schmuller, BPB Publications, 1999. 3. Refactoring: Improving the Design of Existing Code, Martin Fowler, Addison-Wesley Professional. 1999. 4. Working Effectively with Legacy Code, Michael C. Feathers. Pearson Education, Prentice-Hall, 2004. 5. UML Distilled, Second Edition: A Brief Guide to the Standard Object Modeling Language, FowlerM.Scott.K.,2000. 6. Time And Synchronization In Executable Uml https://www.modelint.com/time-and-synchronization-in-executable-uml. 7. Discrete, Continuous and Hybrid Petri Nets (David, R. and Alla, H.; 2004), in IEEE Control Systems Magazine, vol. 28, no. 3, pp. 81-84, June 2008, doi: 10.1109/MCS.2008.920445.

II.Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifact
CLO-1	Understand and Develop Object-oriented design models and refine them to reflect implementation details	Cognitive	3	2	A1, Q1, Midterm,
CLO-2	Demonstrate good and bad coding practices to ensure code quality	Cognitive	2	2	A2, Q2, MidTerm, Final Term
CLO-3	Analyze the major activities within the OO software development process to encourage modularity and reusability.	Cognitive	4	3	A3, Q3, Final term

III.Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV.Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Software construction fundamentals, Software deployment processes, Software engineering process infrastructure, Software engineering process improvement, Systems engineering life cycle models, Evolution processes and activities, Basic concepts of evolution and maintenance, Working with legacy systems, Refactoring, Error handling, exception handling, Apply a wide variety of software construction techniques and modeling, including clean code concepts, state-based and table-driven approaches to low-level design of software, configuration management. Martin Fowler's refactoring concepts and their application to small projects. legacy code concepts. Exception handling, Test driven development.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO1	Course Outline, Assessment Criteria, Introduction to Construction, Key decisions of Construction and software development aspects	Chapter 1,4 [Book 2] Chapter 1 [Book 4]



2		Static Behavioral model VS. Dynamic behavioral model. Context Model, Interaction Model and Model driven engineering Systems Models in Context of Software Construction, Synchronization and timing in UML modeling	Chapter 5 [Book 4]
3		UML Models (Package, Object)	Chapter 14 [Ref Book 2] Chapter 6 [Ref Book 5]
4		UML Models (Communication and Timing)	Chapter 12,17 [Ref Book 5]
5		Introduction of Petri Nets, understanding through examples. Petri Nets vs State machine diagram	Online Source [7]
6		Formation of different Finite Automata, Transition tables Languages, regular expressions and syntax tree	Chapter 2,4,5 [Ref Book 1]
7	CLO2	Clean Code vs Bad code, Writing clean & good code practice, Meaningful Names and Standard Notations	Chapter 1,2 [Book 1]
8		Good Comments: Legal comment, clarification, explanation of intent, Amplification, informative comments	Chapter 4 [Book 1]
9	Midterm Exam		
10		Introduction to formatting in Clean Code .Introduction of Vertical Formatting. Introduction of horizontal formatting and horizontal openness	Chapter 5 [Book 1]
11		Refactoring (First make it work, then make it Right, Conclusion)	Chapter 16, [Book 1]
12		Smells and Heuristics (Comments, Functions, General)	Chapter 17 [Book 1]
13	CLO 3	Test driven development for measuring the software quality	Chapter 9 [Book 1]
14		Software Evolution and its impacts, Software Configuration management, Legacy systems and Michal Feather “Legacy” code concepts	Chapter 9 [Book 4] Chapter 1,2 [Ref Book 4]
15		Criteria of object orientation and Modularity, Approaches to reusability.	Chapter 2, 3 [Book 3] Chapter 16 [Book 4]
16		Error handling core keywords and discussion.	Chapter 7 [Book 1]